Software Engineering and Project Management

Deliverable D1 — Group Agreement

Fall 2026

Team Members

- Ahad Ali Saeed
- Sufyan Ahmad
- Abdullah Ayub

1. Who We Are

All three team members bring frontend and backend development experience along with prior project work, giving the team broad, overlapping technical coverage rather than narrow specialists. This means any member can pick up either side of the stack, which is what makes the fluid role rotation described in Section 4 workable.

Ahad Ali Saeed: Frontend and backend experience; has delivered prior software projects.

Sufyan Ahmad: Frontend and backend experience; has delivered prior software projects. Also taking on the Team Lead / Coordinator role for this project.

Abdullah Ayub: Frontend and backend experience; has delivered prior software projects.

2. Individual Interest in the Course

Ahad Ali Saeed: Wants to learn how to plan and run a software project properly, not just write code. His main interest is in understanding how requirements are gathered and how different parts of a system are integrated into a working whole.

Sufyan Ahmad: Interested in the project management side of software development. Taking on the Team Lead role is a deliberate choice to get hands-on experience with coordinating a team, managing a backlog, and keeping a project on track across multiple sprints.

Abdullah Ayub: Wants to understand how good software design decisions are made and documented. His focus is on learning how to specify requirements clearly and how to make design choices that hold up as a project grows.

3. Target Grade

The team is aiming for Grade A on the project component. This is a public, agreed team commitment: it sets the bar the team will hold itself to when deciding how much effort scrum artefacts, design documentation, and presentation quality deserve, not just the working software itself.

4. Roles and Responsibilities

Per the course's advice, the team is explicitly NOT splitting into a permanent "developers" group and a "documenters" group. Sufyan Ahmad takes on the Team Lead / Coordinator role for the duration of the project — arranging meetings, keeping the Trello board current, tracking deadlines, and acting as the point of contact with the Scrum Coach and Project Owner. This coordination role is fixed because the team needs one stable point of contact, but Sufyan will rotate through frontend/backend/integration work like everyone else when not doing coordination tasks.

Beyond coordination, roles are fluid. All three members will rotate across frontend, backend, and integration work over the course of the project, rather than each person permanently owning one layer.

Member	Primary focus this rotation	Responsibilities
Sufyan Ahmad	Team Lead / Coordinator (fixed for the project) + rotates through dev work	Schedules and runs meetings, keeps Trello board current, tracks deadlines, liaises with Scrum Coach and Project Owner, makes final call if the team is deadlocked.
Ahad Ali Saeed	Rotates: Frontend / Backend / Integration	Whatever the current sprint role requires (see rotation policy below), plus documentation for the components he builds.
Abdullah Ayub	Rotates: Frontend / Backend / Integration	Whatever the current sprint role requires (see rotation policy below), plus documentation for the components he builds.

4a. Are Roles Static or Fluid?

Fluid. The team agreed that fixed "coder" vs. "documenter" roles work against the point of the course, so everyone will move between frontend, backend, and integration work.

- Rotation happens at sprint boundaries (i.e., a member does not switch mid-sprint unless something breaks and needs an emergency reshuffle).
- At the start of each sprint planning session, the team agrees together who takes which area for that sprint, based on workload and what each person wants to try next.
- The Team Lead / Coordinator role stays with Sufyan Ahmad for the whole project for continuity, independent of the frontend/backend/integration rotation.

5. Communication and Collaboration Tools

Trello: Sprint backlog, task board, and tracking (as required by the course).

WhatsApp: The team's day-to-day communication channel for quick questions, coordination, and sharing updates outside scheduled meetings.

The team agreed that anything decision-relevant discussed on WhatsApp should be summarized back onto the Trello board or in meeting notes, so decisions aren't lost in chat history.

6. Meeting Rules

Sufyan, as Coordinator, is responsible for scheduling and running meetings. The team's baseline plan:

- A short weekly check-in (~15 minutes) to confirm progress, surface blockers, and adjust the current sprint's task split if needed.
- Longer sprint planning / review sessions timed around the course's own Sprint / Scrum Coach meetings (every 2 weeks).
- Meetings are called with at least 24 hours' notice on WhatsApp, with a one-line agenda.
- If a member cannot attend, they post their status and any blockers on WhatsApp/Trello beforehand.

7. Handling a Free-Rider Situation

The team agreed on an escalating, good-faith process rather than jumping straight to conflict:

- Step 1 — Direct conversation: the Coordinator (or any member) raises the concern with the individual privately and informally first.

- Step 2 — Team discussion: if it continues, it is raised openly in a team meeting and logged as an agreed action item with a clear deadline.
- Step 3 — Scrum Coach escalation: if the issue persists after Step 2, the team brings it to the Scrum Coach, using this D1 document as the agreed record of what was expected and what happened. If the Scrum Coach confirms the pattern, the member may be removed from the team or have their grade contribution adjusted.

7a. Attendance and Contribution Penalties

Missing meetings or failing to deliver agreed tasks is treated as a free-rider trigger and follows the same escalating process above. Specific penalties the team has agreed to:

- Missing a meeting without notice: the member must post a written update on WhatsApp and Trello before the meeting starts. A second unexcused absence in the same sprint is escalated to Step 2.
- Failing to complete an agreed task by the sprint deadline: the member explains the blocker at the next check-in and agrees a new deadline with the team. A repeated pattern (two or more sprints) moves to Step 2.
- Removal from the team: if the problem persists through all three steps and the Scrum Coach agrees, the member can be removed from the team. The remaining members will inform the course facilitator and adjust the workload split accordingly. This D1 document and the Trello history will serve as evidence.

The goal is to resolve issues within the team wherever possible, and only escalate once there is a documented pattern rather than a single missed task.

8. When and How the Team Collaborates

- Weekly check-in (short, ~15 min) plus sprint planning/review aligned to the course's biweekly Scrum Coach cadence.
- Day-to-day coordination happens asynchronously on WhatsApp and Trello, so members aren't blocked waiting for a scheduled meeting.
- Pairing sessions (two members working together, e.g. on integration) are arranged ad hoc via WhatsApp whenever a task needs more than one person.

The team also noted they will jointly review the course's other registration/administrative tasks (e.g., initial questionnaire, Studium sign-ups) at the first meeting to make sure nothing is missed.

9. Initial Tasks

#	Task	Owner	Target
1	Finalize which 2-player game to build and write a one-paragraph pitch	Whole team	Before Session 1
2	Set up Trello board (backlog, sprint columns) and WhatsApp group	Sufyan	This week
3	Draft initial functional & non-functional requirements	Whole team	Before first Scrum Coach meeting
4	Sketch initial game platform / game engine architecture	Whole team	Sprint 1 planning
5	Agree on tech stack (language, frameworks, repo)	Whole team	Before coding starts
6	Set up shared GitHub repo and confirm everyone can build/run it	Ahad & Abdullah	Before Sprint 1

11. User Requirements

Note: the team is currently planning to build a 2-player digital Ludo game. This is a preliminary decision and will be confirmed with the course teacher before Sprint 1. If the game changes, this section will be updated under the amendment process in Section 13.

User requirements describe what the system must do from the player's point of view, without going into technical detail.

- UR-01: Two players must be able to start a new game from the main menu.
- UR-02: Each player must be able to roll a virtual dice on their turn.
- UR-03: The game must move the player's token automatically after the dice is rolled, following standard Ludo rules.
- UR-04: The game must clearly show whose turn it is at all times.
- UR-05: When a token lands on an opponent's token, the opponent's token must return to its home position.
- UR-06: The game must announce the winner when all of one player's tokens reach the finish.
- UR-07: Players must be able to restart or quit the game at any point.

12. System Requirements

System requirements define the technical behaviour and constraints the software must satisfy. These are also preliminary and will be revised once the tech stack and final game choice are confirmed.

12a. Functional Requirements

- SR-F01: The system must enforce turn order so that only the active player can roll the dice.
- SR-F02: The dice roll must produce a random integer between 1 and 6 using a software random number generator.
- SR-F03: The system must validate every token move against the Ludo rule set before updating the board state.
- SR-F04: The system must detect a capture event and reset the captured token to its home position automatically.
- SR-F05: The system must detect a win condition and display the result screen without requiring any player action.
- SR-F06: The system must support restart and exit actions from any game screen.

12b. Non-Functional Requirements

- SR-NF01: The board and token positions must update within 200 ms of a dice roll on the target machine.
- SR-NF02: The game must run on Windows 10 or later without requiring administrator rights to install or run.
- SR-NF03: The codebase must be stored on GitHub with meaningful commit messages so all team members and the Scrum Coach can follow the project history.
- SR-NF04: The UI must be clear enough for a first-time player to understand turn order and token movement without reading any instructions.

13. Amendments to This Agreement

This document may be updated during the course to better reflect how the team actually works, but only with the agreement of all three members. Any change will be logged with a date and a one-line reason at the end of this document.

Signatures / Agreement

By continuing on the project, all three members confirm they have discussed and agreed to the contents of this document:

Name	Signature
Ahad Ali Saeed	
Sufyan Ahmad	
Abdullah Ayub	